SILVER OAK UNIVERSITY

EDUCATION TO INNOVATION

TUTORIAL - 10 | Automation & Regression Analysis

Aim: Analyze black-box testing, white-box testing, integration testing, and system testing on a sample application (YouTube Mobile App).

Module Name: YouTube App - Functional & UI Test Cases

Test Case ID: yt_01
Tester Name: Aman Sevak
Platform / App: YouTube Mobile Application (Android)
Environment: Android device, Stable internet, Logged-in Google account
Description: This report covers black-box functional and UI test cases executed on the YouTube mobile application covering Home feed, Search, Video Playback, Channel pages, History, Settings, and overall navigation/UI consistency.

1. Home Feed & Search Functionality

Scenario ID: 1a

Scenario: Open YouTube Home Page / Feed

Test Steps:
1. Launch YouTube App
2. Observe the Home tab feed and bottom navigation

Screenshots:

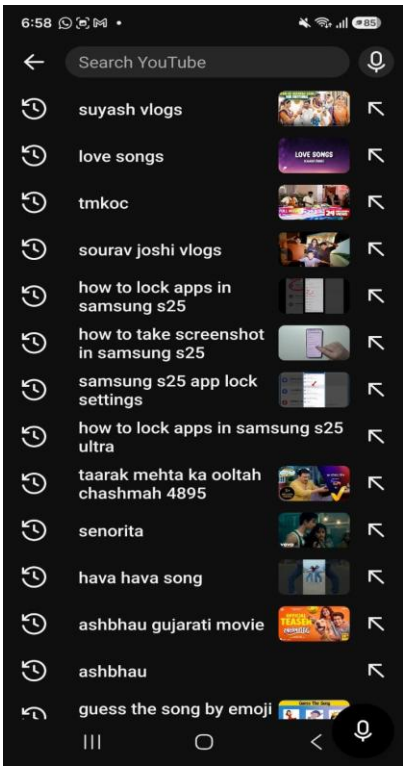


Scenario ID: 1b

Scenario: Search for a keyword (tmkoc / related)

Test Steps:
1. Tap Search icon
2. Enter keyword 'tmkoc' or related
3. Observe suggestions and recent searches

Screenshots:



2. Video Playback & Controls

Scenario ID: 1d

Scenario: Play a video (TMKOC Full Movie example)

Test Steps:

1. Select a video from search or recommendations
2. Tap to play and observe player

Screenshots:



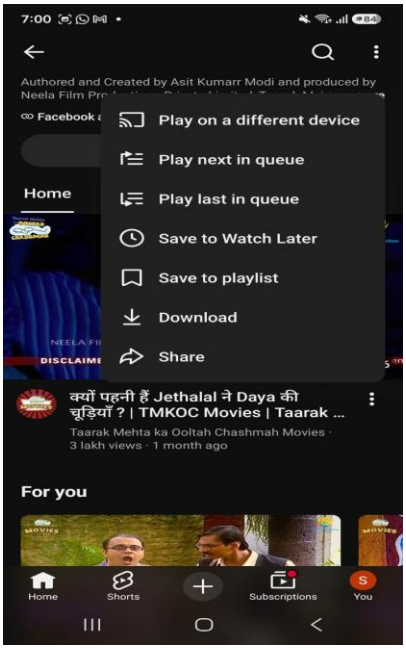
Scenario ID: 1e / 1f

Scenario: Verify player controls & Options menu (3-dot)

Test Steps:

1. While video playing, check play/pause, seek, CC
2. Tap three-dot menu and observe options (Download, Share, Save to Watch Later, Play on different device etc.)

Screenshots:





3. Channel Page & Subscription

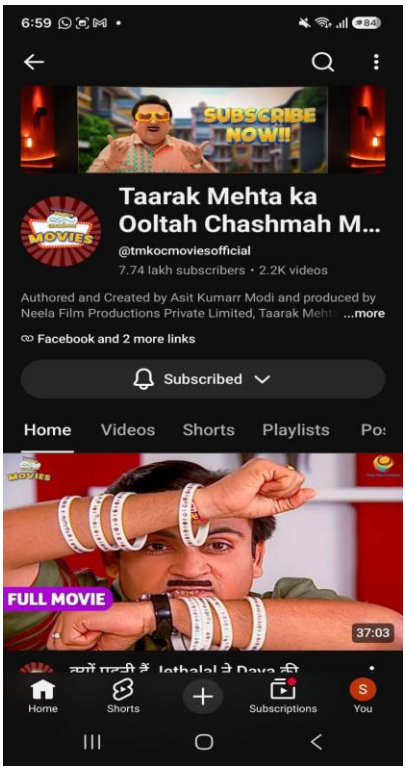
Scenario ID: 2a / 2b

Scenario: Navigate to Channel page and check Subscribe state

Test Steps:

1. From video, tap channel name
2. Observe banner, subscriber count, Subscribe/Subscribed button, video list

Screenshots:



4. History & Settings

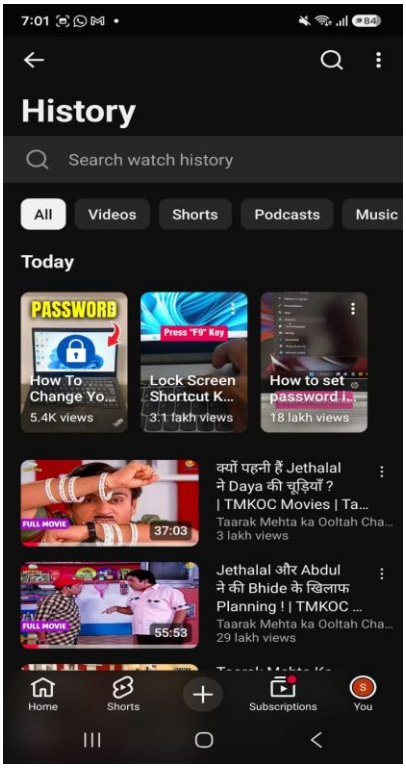
Scenario ID: 3a / 3b

Scenario: Open History page and filter options

Test Steps:

1. Go to You tab → History
2. Observe Today section and filter chips (All, Videos, Shorts etc.)

Screenshots:



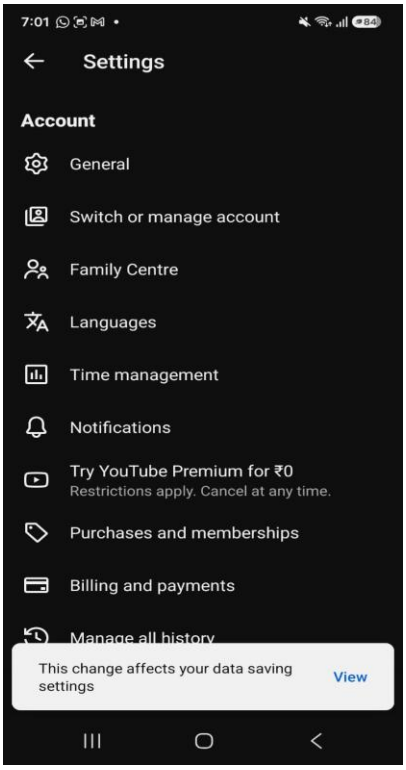
Scenario ID: 3c

Scenario: Open Settings page

Test Steps:

1. From profile/You → Settings
2. Observe Account options: General, Notifications, Languages, Premium etc.

Screenshots:



5. UI / Navigation / Responsive Checks

Scenario ID: 4a

Scenario: Verify bottom navigation bar

Test Steps:

1. Observe bottom nav on Home, Shorts, Subscriptions, You screens
2. Confirm 5 icons: Home, Shorts, Create (+), Subscriptions, You

Screenshots:

[Screenshot attached / Refer Excel for details]

Scenario ID: 4b / 4c / 4d

Scenario: UI alignment, Dark theme contrast, Mobile responsiveness

Test Steps:

1. Check spacing and alignment on video page and lists
2. Navigate in dark mode for contrast
3. Confirm no cut-off or overlapping elements on mobile viewport

Screenshots:

[Screenshot attached / Refer Excel for details]

Summary Of Test Results

Total Test Cases: 22

Passed: 22

Failed: 0

Pass %: 100%

Conclusion and Recommendations

Pass Rate: 100%. The YouTube Mobile App performed well across all tested black-box functional scenarios covering Home feed, Search (with history & suggestions), Video playback & player controls, Options menu (Download / Share / Watch Later), Channel navigation & subscription state, History page with filters, Settings navigation, bottom navigation bar, and UI consistency in dark mode.

Issues Found:

No critical or major functional defects were observed in the tested flows. All core features behaved as expected.

Minor Observation: Sponsored content (e.g., Google Gemini promo) appears on the feed and is clearly labeled. Download of certain content may prompt for YouTube Premium.

Recommendations:

1. Continue monitoring ad frequency so that it does not degrade user experience.
2. Ensure download eligibility messaging is clear for free vs Premium content.
3. Maintain consistent dark-mode contrast across all new feature screens.

Testing Techniques Applied (as per Aim)

Black-box Testing: Primary technique used. Validated inputs (search keywords, menu selections, navigation taps) against expected outputs (feed load, suggestions, player start, channel page, history list) without knowledge of internal source code.

White-box Testing: Limited applicability because application source code is not available. Observed UI state transitions and control flow at the interface level only.

Integration Testing: Verified seamless interaction between modules – Search → Video Player → Channel page → History, and You tab → Settings / History.

System Testing: End-to-end system level validation on real mobile device covering complete user journeys from app launch through content discovery, consumption, and account settings management.

Bug Reports

No major bugs recorded. All 22 test cases passed.

— End of Report —